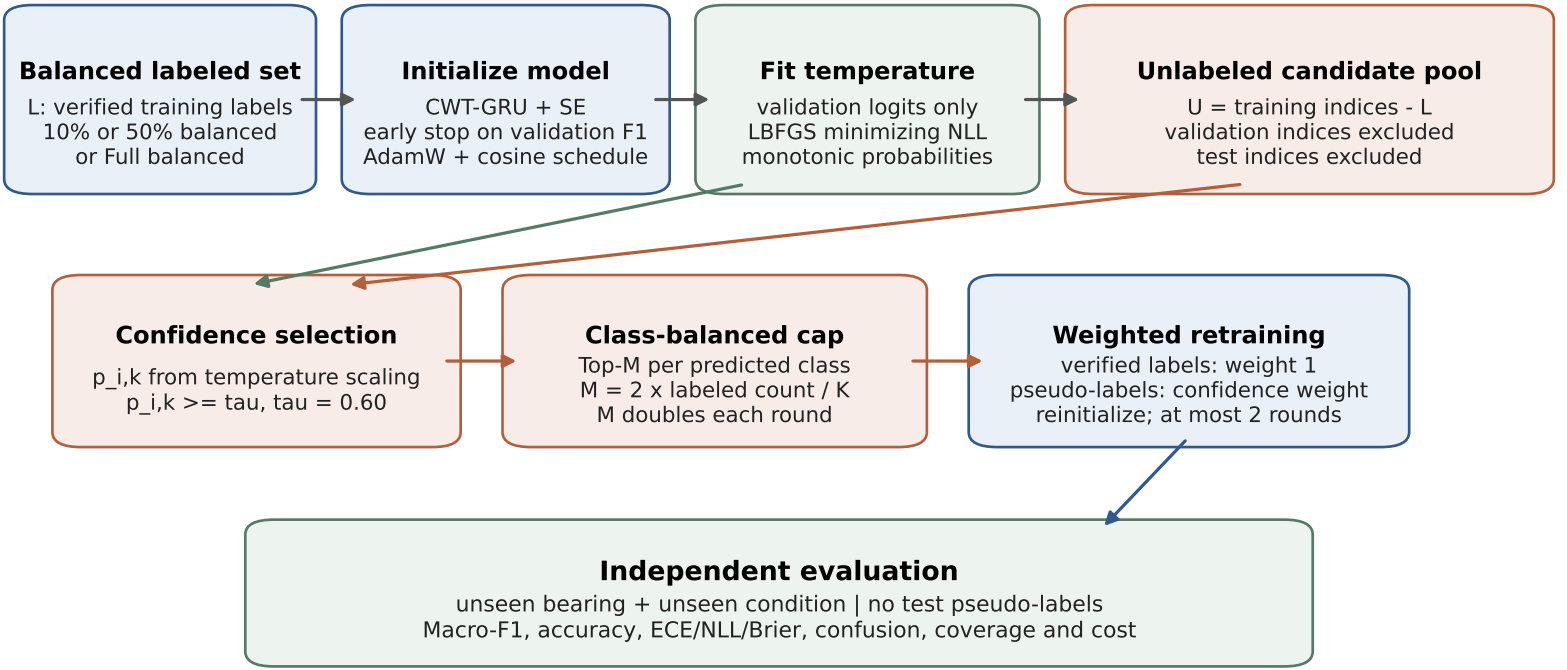
Balanced labeled set
L: verified training labels
10% or 50% balanced
or Full balanced
Initialize model
CWT-GRU + SE
early stop on validation F1
AdamW + cosine schedule
Fit temperature
validation logits only
LBFGS minimizing NLL
monotonic probabilities
Unlabeled candidate pool
U = training indices - L
validation indices excluded
test indices excluded
Confidence selection
p_i,k from temperature scaling
p_i,k >= tau, tau = 0.60
Class-balanced cap
Top-M per predicted class
M = 2 x labeled count / K
M doubles each round
Weighted retraining
verified labels: weight 1
pseudo-labels: confidence weight
reinitialize; at most 2 rounds
Independent evaluation
unseen bearing + unseen condition | no test pseudo-labels
Macro-F1, accuracy, ECE/NLL/Brier, confusion, coverage and cost